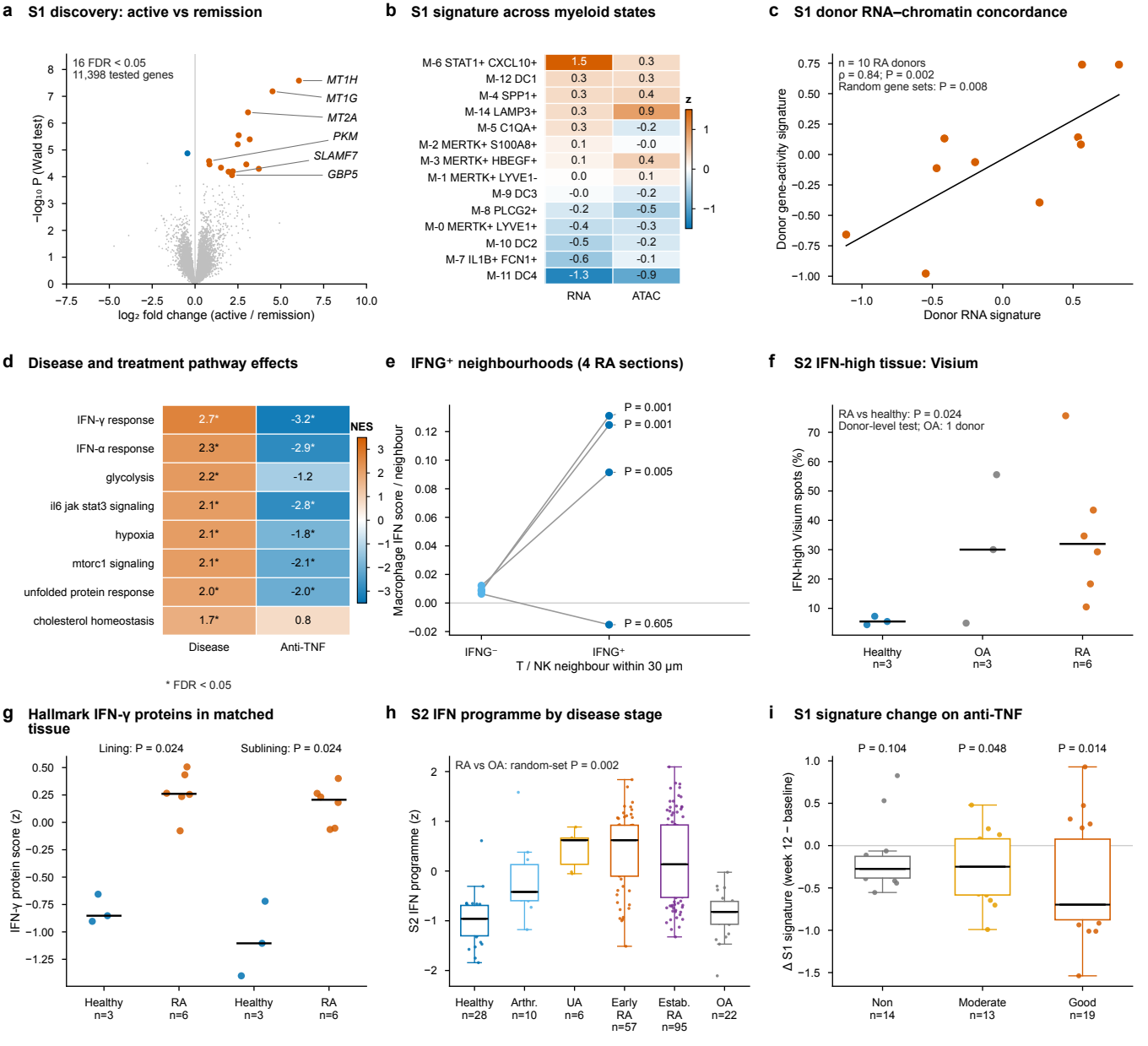

a
S1 discovery: active vs remission
16 FDR < 0.05
11,398 tested genes
MT1H
MT1G
MT2A
PKM
SLAMF7
GBP5
−log10 P (Wald test)
log2 fold change (active / remission)
b
S1 signature across myeloid states
M-6 STAT1+ CXCL10+ 1.5 0.3
M-12 DC1 0.3 0.3
M-4 SPP1+ 0.3 0.4
M-14 LAMP3+ 0.3 0.9
M-5 C1QA+ 0.3 -0.2
M-2 MERTK+ S100A8+ 0.1 -0.0
M-3 MERTK+ HBEGF+ 0.1 0.4
M-1 MERTK+ LYVE1- 0.0 0.1
M-9 DC3 -0.0 -0.2
M-8 PLCG2+ -0.2 -0.5
M-0 MERTK+ LYVE1+ -0.4 -0.3
M-10 DC2 -0.5 -0.2
M-7 IL1B+ FCN1+ -0.6 -0.1
M-11 DC4 -1.3 -0.9
RNA
ATAC
z
c
S1 donor RNA–chromatin concordance
n = 10 RA donors
ρ = 0.84; P = 0.002
Random gene sets: P = 0.008
Donor gene-activity signature
Donor RNA signature
d
Disease and treatment pathway effects
IFN-γ response 2.7* -3.2*
IFN-α response 2.3* -2.9*
glycolysis 2.2* -1.2
il6 jak stat3 signaling 2.1* -2.8*
hypoxia 2.1* -1.8*
mtorc1 signaling 2.1* -2.1*
unfolded protein response 2.0* -2.0*
cholesterol homeostasis 1.7* 0.8
Disease
Anti-TNF
NES
* FDR < 0.05
e
IFNG+ neighbourhoods (4 RA sections)
P = 0.001
P = 0.001
P = 0.005
P = 0.605
Macrophage IFN score / neighbour
IFNG−
IFNG+
T / NK neighbour within 30 µm
f
S2 IFN-high tissue: Visium
RA vs healthy: P = 0.024
Donor-level test; OA: 1 donor
IFN-high Visium spots (%)
Healthy n=3
OA n=3
RA n=6
g
Hallmark IFN-γ proteins in matched tissue
Lining: P = 0.024
Sublining: P = 0.024
IFN-γ protein score (z)
Healthy n=3
RA n=6
Healthy n=3
RA n=6
h
S2 IFN programme by disease stage
RA vs OA: random-set P = 0.002
S2 IFN programme (z)
Healthy n=28
Arthr. n=10
UA n=6
Early RA n=57
Estab. RA n=95
OA n=22
i
S1 signature change on anti-TNF
P = 0.104
P = 0.048
P = 0.014
Δ S1 signature (week 12 − baseline)
Non n=14
Moderate n=13
Good n=19